

Machine-Verified Experimental Mathematics: Four Ramanujan-Style Case Studies

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Abstract

Ramanujan’s method—compute aggressively, conjecture boldly from the data, then seek the identity—produced curiosities that were later revealed as tips of deep icebergs. We ask a *methodological* question: can that method be run today end-to-end with machine-checked rigor, so that “conjecture from computation” terminates in “theorem verified by kernel”? We answer yes through four self-contained case studies: three of the four reach a Lean 4/Mathlib proof, and Case 4 is computation plus paper-proof, resting on the shared formalized matching-polynomial identities. The contribution is *not* any individual theorem—we argue plainly that the four discoveries are, in the grand scheme, minor—but the demonstration that the experimental-to-formal pipeline closes, including (to our knowledge) the first machine-verified instance of the Hall–Puder–Sawin “Ramanujan coverings” circle of ideas, and a small library of reusable formal permanent theory that Mathlib previously lacked. We are equally explicit about the pipeline’s failure modes: one case study is presented as a cautionary tale in which an intermediate “decisive” computation was *retracted* after its proxy invariant was found unfaithful, leaving a razor-sharp but still-open conjecture together with the single experiment that would settle it.

1 Introduction

Ramanujan filled notebooks with identities found by computing far past the point where a pattern becomes “obvious,” then trusting the pattern. A century of hindsight shows how often the obvious pattern hid real structure—partitions and modular forms, the mock theta functions and mock modular forms. The working method was: *compute, conjecture, and only then prove*.

Two things have changed since. First, “compute” can now mean tens of billions of exact operations, pushing the empirical frontier of a combinatorial sequence far enough that the eventual asymptotic regime is actually visible. Second, “prove” can now mean *machine-checked*: a proof assistant’s kernel certifies the argument against a tiny trusted core. This paper is an experiment in running Ramanujan’s loop with both upgrades in place, on four genuinely new

*All data, scripts, and the Lean 4 development are in the `rama-notes` repository; the formalization is the library `RamaLean/` (Lean 4 + Mathlib, standard axioms `[propext, Classical.choice, Quot.sound]`; plus `Lean.ofReduceBool` for the `native_decide` computational cross-checks (Cases 1–2), no `sorry`. The Paper 3 divisibility theorems and the permanent library are on the clean three-axiom base). Correspondence: vicobonfioli@gmail.com.

(if modest) objects, and reporting honestly on what the discipline of formalization does and does not buy.

Our thesis is deliberately narrow. We do *not* claim the four objects below are deep; Section 7 argues the opposite. We claim that the *pipeline*

compute $\rightarrow$ conjecture $\rightarrow$ structural proof $\rightarrow$ Lean formalization $\rightarrow$ decisive experiment when theory stalls

closes end-to-end on real examples, that closing it surfaces reusable infrastructure, and that it imposes a useful honesty: a formalized theorem cannot be quietly wrong, and—as Case 3 shows—a computation engineered to settle a conjecture can force a *retraction* rather than a confirmation.

2 The pipeline

Two features distinguish the loop from ordinary formalization. (i) The conjectures are *discovered* here, not imported from the literature to be re-proved. (ii) When a structural proof stalls, we engineer a *decisive computation* pushed to the feasibility frontier (Gray-code Ryser for permanents; sieve methods to 10^7 for partitions), and we allow that computation to *refute*. The formalization step is not decoration: it is where an argument that “reads correctly” is forced to be correct, and several of our human-written first drafts did not survive it.

3 Case study 2: random lifts of cycles, machine-verified

Let C_n be the n -cycle and consider a uniformly random permutation r -lift (an r -sheeted covering given by assigning a permutation in S_r to each edge). The expected characteristic polynomial of the “new” part of the lift—the part orthogonal to the base spectrum—is, by the Hall–Puder–Sawin theory of Ramanujan coverings [1],

$$\Phi_{C_n,r}(x) = \chi_{C_n}(x) \cdot U_{r-1}(T_n(x/2)), \quad (1)$$

where T_n, U_{r-1} are Chebyshev polynomials of the first and second kind. The quotient $U_{r-1}(T_n(x/2))$ is the $(r-1)$ -matching polynomial of C_n ; its real-rootedness is the finite- r generalization, due to [1], of the Heilmann–Lieb theorem and of the interlacing method of Marcus, Spielman, and Srivastava [2]. This structure was established in 2018; we do not improve it. What we contribute is threefold.

An independent elementary derivation. We obtain (1) through a holonomy/transfer-matrix computation and the exponential formula, rather than through matching-polynomial combinatorics, yielding en route the generating function $\sum_{r \geq 0} \Phi_{C_n,r}(x) z^r = (1-z)^2/(1-2Yz+z^2)$ with $Y = T_n(x/2)$.

An exact, effective spectral edge. Whereas [1] give the asymptotic Ramanujan bound (largest new eigenvalue $\rightarrow 2$), we determine the edge exactly at every finite r : the largest root of $U_{r-1}(T_n(x/2))$ is

$$x^* = 2 \cos\left(\frac{\pi}{nr}\right), \quad \text{so the spectral gap is } 2 - 2 \cos\left(\frac{\pi}{nr}\right) \sim \frac{\pi^2}{(nr)^2}.$$

Full formalization. Both the factored form (1) (over a general commutative ring, with `native_decide` cross-checks) and the spectral edge are machine-checked in `RamaLean/Paper2*.lean`. The edge is the theorem

`Paper2SpectralEdge.spectral_edge_is_root`: $U_{r-1}(T_n(\cos(\pi/(nr)))) = 0$ for $0 < n, 2 \leq r$,

proved from $T_n(\cos \theta) = \cos n\theta$ and the vanishing of U_{r-1} at $\cos(\pi/r)$. To our knowledge this is the first formally verified statement inside the Ramanujan-coverings circle.

4 Case study 3: the 2-adic gcd permanent (a cautionary tale)

Let $a(n) = \text{per}[\text{gcd}(i, j)]_{1 \leq i, j \leq n}$. The determinant analogue is Smith’s 1876 identity $\det[\text{gcd}(i, j)] = \prod_{k \leq n} \varphi(k)$; the permanent is catalogued as OEIS A085244 (2003), but its arithmetic (divisibility, 2-adic growth) appears not to have been studied. We establish (all machine-checked): $2, 3, 4, 8 \mid a(n)$ in suitable ranges, and the sharp linear 2-adic growth rate

$$v_2(a(n)) \geq n - 2\lfloor \log_2 n \rfloor - 2 \quad (\text{“growth constant } c = 1\text{”}), \quad (2)$$

the theorem `Paper3C1.two_pow_dvd_permanent_c1`. The proof rests on four new, reusable permanent-divisibility lemmas (Section 6).

The discovery, and the retraction. Exact Gray-code Ryser values of $v_2(a(n))$ at the “peaks” $n = 2^k + 1$ yield $v_2(a) = 3, 5, 11, 25$ for $k = 2, 3, 4, 5$, i.e.

$$v_2(a(2^k + 1)) = 2^k - 2k + 3 \quad \Longleftrightarrow \quad D(2^k + 1) := v_2(n!) - v_2(a) = 2k - 4 \sim 2 \log_2 n, \quad (3)$$

a clean line through all four exact points, suggesting the deficit D is *unbounded*, of order $\Theta(\log n)$, against the $O(\log n)$ ceiling implied by (2). Full Ryser at the next peak $n = 65$ is 2^{65} operations—out of reach—so we engineered a factored proxy $N_0 = \sum_v \text{per}(A_v) \text{per}(B_v)$ (the weight-0 grade of the 2-adic decomposition $a = \sum_w 2^w N_w$), computable at $n = 65$, and obtained $v_2(N_0(65)) = 58$.

We initially read this as *refuting* (3) (it predicts $v_2(a(65)) = 55$). A validation check then showed the proxy is unfaithful: $v_2(a) - v_2(N_0)$ is not identically 0 but $-2, +1, +3$ at $n = 7, 12, 13$. Hence $v_2(N_0(65))$ does *not* determine $v_2(a(65))$, and the “refutation” was withdrawn. What remains is a genuine, sharply-posed tension:

- the closed form (3) fits 4/4 exact peaks (including $k = 2$, where the proxy mechanism below fails), and predicts $D(65) = 8$ (unbounded);
- at $k = 3, 4, 5$ the weight-0 grade dominates ($v_2(a) = v_2(N_0)$), but the grade-profile margins run 0, 1, 3, 1 (over $k = 2, 3, 4, 5$)—fluctuating, not growing, so this dominance is fragile; *if* it nonetheless persisted, $v_2(N_0(65)) = 58$ would force $D(65) = 5$ (a bounded “turnover”).

These are incompatible at $k = 6$: exactly one pattern breaks there. Attacking the equivalent recursion $v_2(a(2^k + 1)) = v_2(a(2^{k-1} + 1)) + (2^{k-1} - 2)$ shows it is *not* an independent, easier target—it holds iff a low grade overtakes $w = 0$ at $k = 6$, iff the deficit is unbounded. What *is* unconditional is the upper bound $D(2^k + 1) \leq 2k$ ($k \geq 3$), from (2); a matching growing lower bound—equivalently the exact conjecture (3)—is open. A formalized closed-form fact about the odd- φ sub-sum ($v_2(A(2^k + 1)) = v_2(n!) - 1$, `Paper3Atom1`) gives $D \geq 2$ infinitely often *conditional* on the empirically-clear but unproven $v_2(B) < v_2(A)$. The single decisive feasible experiment is the exact $n = 33$ grade profile (a Gray-code Ryser over $(\mathbb{Z}/2^{64})[t]$,

`code/rust_grade/`); its margin between the $w = 0$ grade and its nearest competitor discriminates the two patterns.

We keep this case in the paper precisely because it is where the method behaved best as *method*: it produced a falsifiable conjecture, an intermediate result, the discovery that the intermediate result’s proxy was unfaithful, a clean retraction, a proof that the natural next target is equivalent to the original, and an explicit pointer to the one computation that would decide it. The formalized core (2) is untouched by any of this.

5 Case studies 1 and 4 (brief)

Case 1: partition self-divisibility. We study $n \mid p(n)$ -type divisibility of the partition function, computed to 10^7 . A probabilistic $\approx \sum 1/n$ heuristic (divergent) predicts infinitely many hits; one supporting proposition is formalized. This is the shallowest case: the varying modulus breaks the modular-forms machinery that gives Ramanujan’s classical congruences their depth, so it stays an honest curiosity.

Case 4: d -matching polynomial coefficients. Inside the same HPS object of Case 2, the coefficients $c_k(d)$ of the d -matching polynomial are polynomials in graph invariants; the two leading coefficients are *universal* (functions of $|E|$ and the path count P only), with c_4 computed exactly on eight graphs. A modest structural refinement of a known object.

6 The formalized library

Mathlib had no permanent-divisibility results and no permanent Laplace expansion. Closing the pipeline produced the following, each a candidate for upstreaming:

1. `permanent_succ_column_zero`: Laplace/cofactor expansion of the permanent (Mathlib had this only for `det`; the sign-free proof is shorter).
2. `two_pow_dvd_permanent_odd`: an all-odd matrix has $v_2(\text{per}) \geq \text{size} - \log$.
3. `two_factorial_dvd_permanent`: two identical row-groups give $|A|!|B|! \mid \text{per}$, via a free $\text{Sym}(A) \times \text{Sym}(B)$ action.
4. `two_pow_dvd_permanent_zeroed`: a zeroed-corner odd matrix has $v_2(\text{per}) \geq \text{size} - 2 \log$.
5. `card_dvd_sum_of_free_invariant`: a free finite-group action with an invariant summand gives $|G| \mid \sum$.
6. `spectral_edge_is_root` (Section 3).

The permanent lemmas share one idea—divisibility of a permanent by the order of a free group acting on the index set while preserving the summand—which we isolate as (5) and reuse throughout.

7 Honest assessment

We state the limitations plainly, because the paper’s credibility depends on it. Individually the four discoveries are minor (on a grand-scheme scale we would rate them 2–4 out of 10): they

concern niche objects, and where they touch deep structure—Case 2, the Ramanujan-coverings framework—that structure was established by others in 2018, capping the novelty. Rigor and machine-checking do *not* buy significance; a formally verified statement about a shallow object is still shallow. Case 3’s central number-theory question is open, and our own analysis suggests its natural proof targets are equivalent to it rather than easier.

The claim is therefore methodological. What the pipeline delivers that a pen-and-paper note does not is: (a) a reproducible, kernel-checked realization of the compute-conjecture-prove loop, end to end, on discovered (not imported) objects; (b) the first formally verified instance of the Ramanujan-coverings circle of ideas; (c) a reusable formal permanent library Mathlib lacked; and (d) an enforced discipline of retraction when a computation’s proxy proves unfaithful. Not a notebook of new deep mathematics—a proof of concept that Ramanujan’s method scales to *rigor*, if not yet to *depth*.

References

- [1] C. Hall, D. Puder, W. F. Sawin, *Ramanujan coverings of graphs*, Adv. Math. **323** (2018), 367–410.
- [2] A. Marcus, D. Spielman, N. Srivastava, *Interlacing families I: bipartite Ramanujan graphs of all degrees*, Ann. of Math. **182** (2015), 307–325.
- [3] O. J. Heilmann, E. H. Lieb, *Theory of monomer-dimer systems*, Comm. Math. Phys. **25** (1972), 190–232.
- [4] H. J. S. Smith, *On the value of a certain arithmetical determinant*, Proc. London Math. Soc. **7** (1875–76), 208–212.
- [5] H. J. Ryser, *Combinatorial Mathematics*, Carus Math. Monographs **14**, MAA, 1963.
- [6] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.